Kaggle uses cookies from Google to deliver and enhance the quality of its services and to analyze traffic. Learn more OK, Got it.

Earth-Space AI Research (ESAIR) Lab · Community Prediction Competition · 23 days to go

Solar Filament Segmentation Challenge 2026

Automatic Segmentation of Solar Filaments using Computer Vision algorithms (with or without fitting) from GONG's H-Alpha Observations

Submit Prediction ···

Dataset Description

For this competition, you will need (1) images and (2) ground-truth annotations. Both of these items can be downloaded in this page. Below, click on the "Download All" button, and then choose "Download as zip". When the download is complete, unzip the downloaded file. The content of the unzipped file is shown below:

```
MAGFiLO_1.0_Kaggle_2026/
|-- train/
|   |-- train_images/
|   |-- MAGFiLO_1.0_Annotations_kaggle2026_train.json
|-- test/
    |-- test_images/
```

Images and Annotations

The goal is to identify solar filaments from solar images. Solar filaments are best visible in H-Alpha images. Therefore, we use the observations (images) captured by H-Alpha instruments of Global Oscillations Network Group (GONG). These are the same images used for creating the manual annotation of solar filaments available in MAGFiLO.

These H-Alpha observations are images of size `2048 x 2048` pixels, which, for the sake of this competition, have been converted from their original FITS format to the standard 8-bit JPEG format. Contestants should note that these

images are grayscale and should not be processed as color images (e.g., RGB images).

Each observation's capture time and instrument are encoded in its file name as follows: `YYYYMMDDHHMMSSII`. For example, `20260901165702Bh.jpeg` corresponds to the capture time of September 1, 2026, at 16:57:02, using the Big Bear (Bh) Solar Observatory.

The H-alpha observations in MAGFiLO often contain multiple filaments. Further, all filaments present within one image may be annotated multiple times by different annotators, independently. As an example, the image with `file_name` as `20260901165702Bh.jpeg` may have two sets of annotations, e.g., `010101-20260901165702Bh` and `010102-20260901165702Bh` (note that they are different by only one digit). You may treat those images simply as "different" images.

For this competition, the ground-truth data are the filament **segmentations**. For storage efficiency, instead of binary masks, we store segmentations as polygons in RLE format. The conversion (RLE <-> Binary Masks) are lossless.

The annotations are structured into the COCO-style data format. That means, you can take advantage of many common operations using the `pycocotools` python package (see https://github.com/ppwwyyxx/cocoapi).

More Info about JSON File

```
{
"info": info,
"images": [image],
"annotations": [annotation],
"licenses": [license],
"categories": [category],
}

info{
"year": int,
"version": str,
"description": str,
"contributor": str,
"url": str,
"date_created": datetime,
}

license{
"id": int,
"name": str,
"url": str,
}

categories[
{
   "id": int,
   "name": str,
   "supercategory": str,
```

```
}]

annotation{
"id": str,
"image_id": int,
"category_id": int,
"segmentation": [[]],
"area": float,
"spine": [],
"bbox": [x, y, width, height],
"iscrowd": 0 or 1,
}

image{
"id": str,
"width": int,
"height": int,
"file_name": str,
"license": int,
"date_captured": datetime,
}
```

- **info:** corresponds to a dictionary containing metadata description of the dataset.
- **images:** corresponds to a list of image dictionaries.
- **annotations:** corresponds to a list of annotation dictionaries.
- **licenses:** corresponds to a dictionary containing the license information of the GONG H-Alpha images.
- **categories:** corresponds to a list of category dictionaries.
- **image:** corresponds to a dictionary containing information about an annotated image, including its name and downloadable URL.
- **annotation:** corresponds to a dictionary containing information about an annotated filament, including its segmentation and bounding box.
- **license:** corresponds to a dictionary describing the license of an image.
- **category:** corresponds to a dictionary containing the category information that each filament may be described with.

More Information on Annotations

- A bounding box in `annotation["bbox"]` is represented with a list of 4 values, `[x, y, width, height]` , where `x,y` corresponds to the top-left corner of the box. All values are in pixels unit.
- A segmentation in `annotation["segmentation"]` is a list of floats representing a closed path. Each path forms a polygon capturing a filament's shape. A polygon made with n points is represented as `[x_0, y_0, x_1, y_1, ..., x_(n-1), y_(n-1)]` where `x_0 = x_(n-1)` and `y_0 = y_(n-1)` .
- The area in `annotation["area"]` is the segmentation area (not the bbox area). This is computed using pycocoapi package, by first converting the

polygon into RLE.

- A spine in `annotation["spine"]` is a list of floats representing a path. Each path captures a filament's spine. Disconnected line segments are not allowed. A spine with `n` points is represented as `[x_0, y_0, x_1, y_1, ..., x_(n-1), y_(n-1)]`.
- The tag `annotation["iscrowd"]` is always zero indicating that each segmentation corresponds to a single filaments (i.e., no group of filaments is annotated together).
- Although `annotation["segmentation"]` is a list, it only contains one polygon. That is, each filament is annotated using a single-piece polygon.
- The class labels listed in `category["name"]` are: "Left" (id = 1), "Right", (id = 2), "Unidentifiable" (id = 3), and "Ambiguous" (id = 4).
- The image ids in `image["id"]` is a string that shows the annotator's batch name, as well as the name (together, unique). For example, id `010401-20160920230134Lh` indicates that the image with name `20160920230134Lh.jpg` is annotated by the annotator `010401`, and it might have also been annotated by two other annotators, `010402` and `010403`, in the same group.
- The annotation ids in `annotation["id"]` is a string (e.g., `a7639f8a-c76d-43a7-b392-e75842262b75`) generated by the annotation platform, which is unique for filaments.

MAGFiLO_1.0_Kaggle_2026 (2 directories)

test	train
1 directories	1 directories, 1 files

Metadata

License

Attribution-NonCommercial 4.0 International (CC BY-NC 4.0)